

CyLab Challenge: AI Foundations I

Date: 2026-08-08 | **Category:** AI/ML | **Difficulty:** Easy | **Time:** 3 hours

The Challenge

A beginner-friendly journey through perceptrons—the building blocks of neural networks. The challenge reconstructs AI fundamentals by exploring how neurons work, how perceptrons are trained, and why responsible AI use matters.

What I Actually Did

- Reviewed perceptron basics and refreshed knowledge of weights/biases
- **What I found:** My understanding was superficial—knew terms but not deep mechanics
- Experimented with different weight/bias values
- Observed how learning rate impacts convergence
- Debugged when model failed to converge
- Applied mathematical concepts to achieve desired output

What I Learned

Technical Skills:

- Perceptron fundamentals & training mechanics
- Deep understanding of weights, biases, and learning rates
- AI is mathematical—understanding the math is essential

AI Security Insights:

- Knowing what a model *cannot* do is as important as what it can
- Limitations lead to vulnerabilities (adversarial attacks, data poisoning)
- Responsible AI is built into every decision, not an add-on

Screenshots/Proof of work

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whether the neuron fires or not. Use those responses to figure out the decision boundary, then line up the last eight outputs to read the ASCII for **p (0110000)** and you'll earn the flag. Connect with netcat:

```
$ nc aureolin-pixie.cylabacademy.net 56250
```

The input bounds are shown on connect to keep the search small. You cannot submit the same (x, y) pair back-to-back.

INSTANCE Expires in 10:58 Restart Instance

HINTS

- 1 Reveal hint
- 2 Reveal hint
- 3 Reveal hint

FLAG

Submit

debug info: 2c1100000 at p: 0:772 12000007

Bookmark 1,252 99%

Neuron Meet 2D-0 10:58 RUNNING

```
Perceptron fires!
Recent outputs (3/8): 111
[4/128] (x,y)> 6 7
Perceptron fires!
Recent outputs (4/8): 1111
[5/128] (x,y)> 9 3
Perceptron fires!
Recent outputs (5/8): 11111
[6/128] (x,y)> 10 9
Perceptron fires!
Recent outputs (6/8): 111111
[7/128] (x,y)> 3 4
Perceptron fires!
Recent outputs (7/8): 1111111
[8/128] (x,y)> exit
Goodbye.
harshita@Harshita-MacBook-Pro-8 ~ % nc aureolin-pixie.cylabacademy.net 56250
Welcome to Neuron Meet 2D-0!
Probe the 2D perceptron to coax out the ASCII for 'p'.
Send two numbers (x, y) to see if the perceptron fires (1) or stays quiet (0).
- Bounds: [-10.0, 10.0] for both x and y
- Output rule: w1*x + w2*y + b >= 0 -> 1, else 0.
- No back-to-back repeats of the same (x, y) pair.
- Goal: make the last 8 outputs read 01110000 (ASCII 'p').
- Command: RESET to clear the firing history.
- Format: x,y or x y (comma or space separated)
Type HELP for a reminder or EXIT to quit.

[1/128] (x,y)> 2 1
Perceptron stays quiet.
Recent outputs (1/8): 0
[2/128] (x,y)> 10 2
Perceptron fires!
Recent outputs (2/8): 01
[3/128] (x,y)> 10 6
Perceptron fires!
Recent outputs (3/8): 011
[4/128] (x,y)> 4 6
Perceptron fires!
Recent outputs (4/8): 0111
[5/128] (x,y)> 2 1
Perceptron stays quiet.
Recent outputs (5/8): 01110
[6/128] (x,y)> 1 2
Perceptron stays quiet.
Recent outputs (6/8): 011100
[7/128] (x,y)> 2 1
Perceptron stays quiet.
Recent outputs (7/8): 0111000
[8/128] (x,y)> 1 2
Perceptron stays quiet.
Recent outputs (8/8): 01110000
Pattern matched! ASCII 'p' unlocked. Here is your flag:
academy{2d_n3ur0n_m3t_892056ff}
harshita@Harshita-MacBook-Pro-8 ~ %
```

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Artificial Intelligence Easy
by LT 'yrcat' Jones · AI Foundations I

Play through an AI ethics interactive fiction where confident AI outputs can be wrong, subtly wrong, or almost-right. Verify claims, code, and citations to learn practical habits for safe AI-assisted work. Connect with netcat:

```
$ nc aureolin-pixie.cylabacademy.net 64303
```

INSTANCE Expires in 8:48 Restart Instance

HINTS

- 1 Reveal hint
- 2 Reveal hint

FLAG

Submit

debug info: 2c1000000 at p: 0:783 12000002

Bookmark 3,664 93%

Trust But Verify 08:47 RUNNING

```
"Trust but verify."
----
(Press Enter to continue...)
----

"Exactly," ARIA says.
----
(Press Enter to continue...)
----

=== ENDING ===

The proposal is submitted. Whether the path here was smooth or full of
corrections, Ren has learned something most people in 2631 haven't fully
internalized yet.

----
(Press Enter to continue...)
----

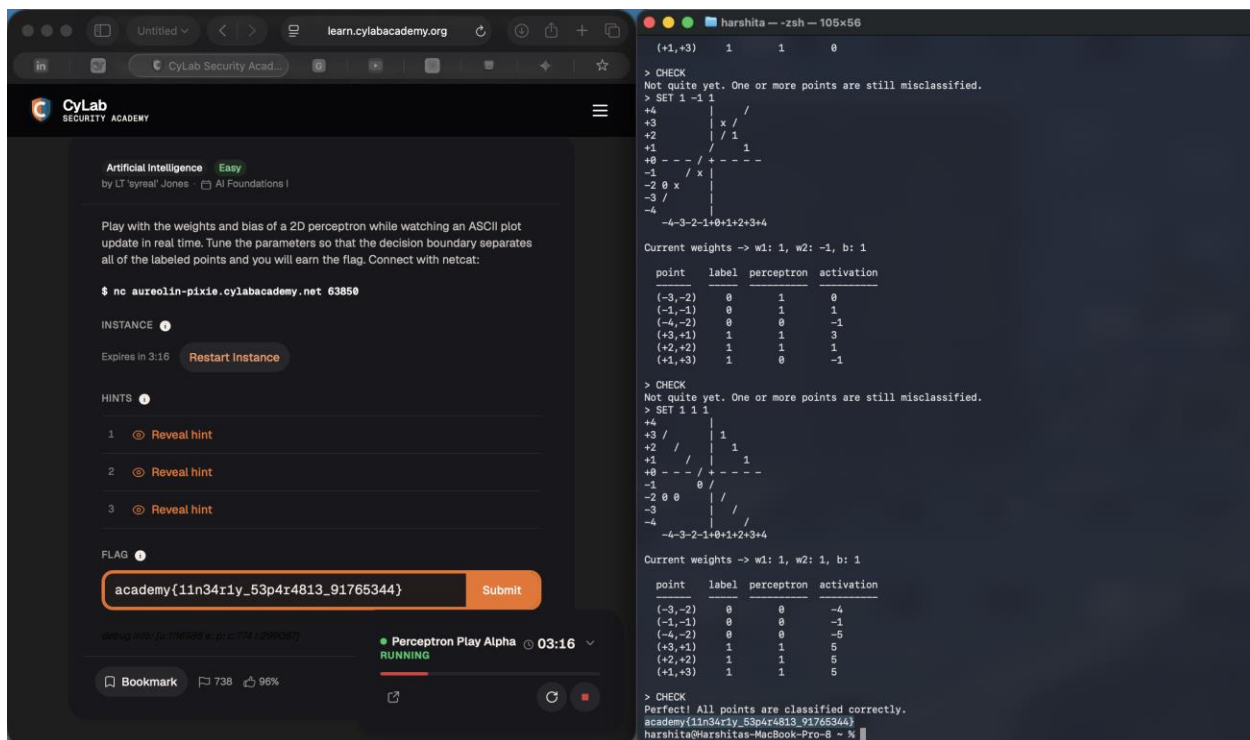
ARIA speaks one last time.
----
(Press Enter to continue...)
----

"I want to be useful to you, Ren. I want to help you think faster, research
deeper, write better. But I need you to be my editor, my fact-checker, my
critical eye. I genuinely don't know when I'm wrong. You have to be the one who
finds out."
----
(Press Enter to continue...)
----

"That's a weird partnership," Ren says.
----
(Press Enter to continue...)
----

"It is," ARIA agrees. "But it's the right one."
----
(Press Enter to continue...)
----

"By the way," ARIA adds, "here's something you can actually verify:
academy{7ru57_15_34rn3d_45944167}"
----
(Press Enter to continue...)
----
```



What I'd Do Differently

I wasted time struggling with concepts I thought I already knew. Many attempts timed out because I jumped in without reviewing fundamentals.

Next time: Review basics thoroughly before starting. The challenge became much easier once I understood the math deeply.

Career Connections

SOC: Understanding model behavior helps detect AI-based attacks and adversarial inputs.

GRC: AI limitations are high-risk findings in security audits. Compliance frameworks (NIST AI RMF, EU AI Act) require understanding AI behavior.

AI Security: This foundation is essential for understanding adversarial ML and data poisoning attacks.

Key Takeaway

"Every AI system has limitations. Understanding what a model can't do is just as important as understanding what it can. This is the foundation of responsible AI security."